

Zamzam jirow

Part A - Project Setup Using the CLI

- * mkdir creates folders.
- * cd moves into a folder.
- * touch creates empty files.
- * echo ... > README.md puts text into a file using output redirection.
- * ls displays files.
- * cat displays file contents.
- * find . -print displays the project contents recursively.

src contains the actual program code, tests contains code used to test the program, and docs contains documentation. Keeping them separate makes a project easier to understand, maintain, and expand.

```
/tests
/docs
/docs/README.md
/src
/src/config.py
/src/utils.py
/src/main.py
iohamed@Mohameds-MacBook-Air python_lab %
```

Part B: Git Initialization and First Commit

```

mohamed@Mohameds-MacBook-Air python_lab % git init
Reinitialized existing Git repository in /Users/mohamed/python_lab/.git/
mohamed@Mohameds-MacBook-Air python_lab % printf "__pycache__/\n*.pyc\n.env\n" > .gitignore
mohamed@Mohameds-MacBook-Air python_lab % cat .gitignore
__pycache__/
*.pyc
.env
mohamed@Mohameds-MacBook-Air python_lab % git add .
mohamed@Mohameds-MacBook-Air python_lab % git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   .gitignore
        new file:   docs/README.md
        new file:   src/config.py
        new file:   src/main.py
        new file:   src/utils.py

mohamed@Mohameds-MacBook-Air python_lab % git commit -m"set up python lab project structure"
[master (root-commit) 55bfb5f] set up python lab project structure
Committer: Mohamed <mohamed@Mohameds-MacBook-Air.local>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

5 files changed, 4 insertions(+)
create mode 100644 .gitignore
create mode 100644 docs/README.md
create mode 100644 src/config.py
create mode 100644 src/main.py
create mode 100644 src/utils.py
mohamed@Mohameds-MacBook-Air python_lab % git log --oneline
55bfb5f (HEAD -> master) set up python lab project structure
mohamed@Mohameds-MacBook-Air python_lab % █

```

.gitignore is a file that tells Git which files and directories should not be tracked or included in commits. In this project, it prevents Python cache folders, compiled Python files, and .env files from being added to the repository. This helps keep the repository clean and prevents unnecessary or sensitive files from being uploaded.

The __pycache__/ and *.pyc patterns are important because Python automatically creates cache folders and compiled Python files when programs are run. These files are generated automatically and are not normally needed in the source-code repository. Ignoring them keeps the Git repository organized and prevents unnecessary files from being tracked.

The .env pattern is also important because .env files may contain sensitive information such as passwords, API keys, or other private configuration details.

Git commit history shows how a project has developed over time. Each commit records a set of changes along with information such as the commit identifier, author, date, and commit message. This allows developers to see what changes were made, when they were made, and how the project has progressed from one stage to another.

Part C - Writing and Committing Python Code

utils.py

```
from utils import square, is_even, celsius_to_fahrenheit

number = float(input("Enter a number: "))

print("Square:", square(number))

if is_even(number):
    print("The number is even.")
else:
    print("The number is odd.")

print("Fahrenheit equivalent:", celsius_to_fahrenheit(number))
```

main.py

```
from utils import square, is_even, celsius_to_fahrenheit
```

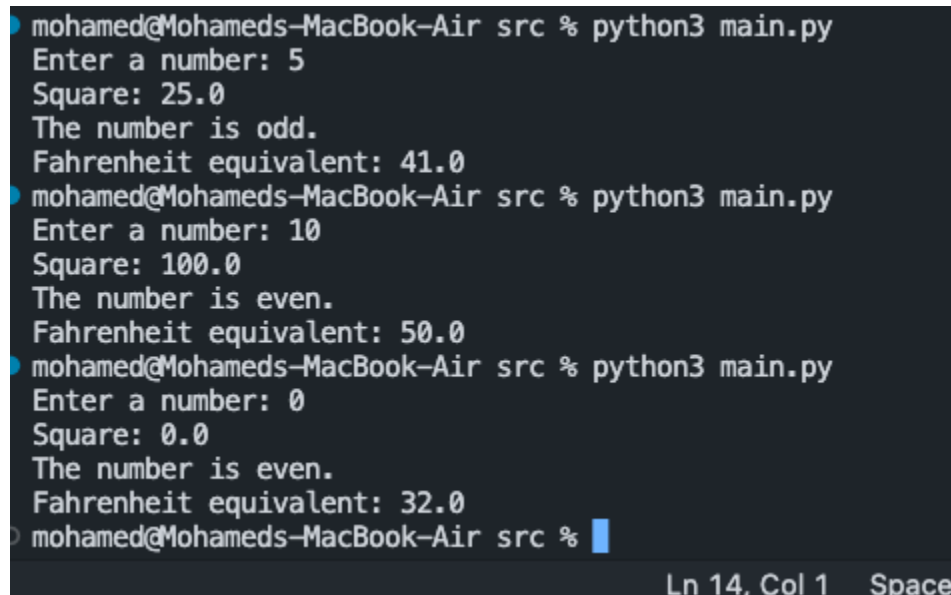
```
number = float(input("Enter a number: "))

print("Square:", square(number))

if is_even(number):
    print("The number is even.")
else:
    print("The number is odd.")

print("Fahrenheit equivalent:", celsius_to_fahrenheit(number))
```

Python's import system allows one Python file to use functions and other code defined in another Python file. In this project, the three functions `square`, `is_even`, and `celsius_to_fahrenheit` are defined in `utils.py`. The statement `from utils import square, is_even, celsius_to_fahrenheit` in `main.py` imports these functions so that they can be called from the main program. This keeps the project organized by separating reusable functions from the main program logic.



```
mohamed@Mohameds-MacBook-Air src % python3 main.py
Enter a number: 5
Square: 25.0
The number is odd.
Fahrenheit equivalent: 41.0
mohamed@Mohameds-MacBook-Air src % python3 main.py
Enter a number: 10
Square: 100.0
The number is even.
Fahrenheit equivalent: 50.0
mohamed@Mohameds-MacBook-Air src % python3 main.py
Enter a number: 0
Square: 0.0
The number is even.
Fahrenheit equivalent: 32.0
mohamed@Mohameds-MacBook-Air src %
```

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Part D - Publishing to GitHub and Branch Workflow

[Code](#) [Issues](#) [Pull requests](#) 1 [Actions](#) [Projects](#) [Wiki](#) [Security and quality](#) [Insights](#) [Settings](#)

add personalized greeting feature #1

 Open **zamzamjirow** wants to merge 1 commit into **main** from **feature/add-greeting** 

 Conversation 0  Commits 1  Checks 0  Files changed 2



zamzamjirow commented 1 minute ago Owner ...

No description provided.



  **add personalized greeting feature** 988de22



 **No conflicts with base branch**
Merging can be performed automatically.

Merge pull request

You can also merge this with the command line. [View command line instructions.](#)

Still in progress? [Convert to draft](#)

 **Add a comment**

Reviewers
No reviews
Still in progress? [Cancel](#)

Assignees
No one—[assign yourself](#)

Labels
None yet

Projects
None yet

Milestone
No milestone

Development
Successfully merged



Your main branch isn't protected

Protect this branch from force pushing or deletion, or require status checks before merging. [View documentation.](#)

Dismiss

Protect this branch



main



2 Branches



0 Tags

Go to file

T

Add file

<> Code



zamzamjirow Delete python lab assignment.pdf

72b6307 · 1 minute ago

7 Commits



docs

set up python lab project structure

last week



src

add python utility functions and main program

last week



.gitignore

set up python lab project structure

last week



config.py

add python lab files to repository root

25 minutes ago



main.py

add python lab files to repository root

25 minutes ago



utils.py

add python lab files to repository root

25 minutes ago



README



my python lab project